

Industrial Supply Sales Performance Dashboard Report

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Data Visualization

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Introduction

In the industrial supply business, looking at visual charts instead of messy spreadsheets helps managers see trends much faster. This project focuses on analyzing bulk sales data for a typical industrial and construction material wholesale distributor. Understanding these sales trends matters because bulk business-to-business (B2B) orders involve large sums of money, meaning any unmonitored drop in a sales channel can severely hurt a company's final profits. By turning raw transaction numbers into simple pictures, we help business leaders quickly see how they are making money and where to grow.

Project Background

An industrial supply distributor typically sells high-value items like structural steel beams, custom metal parts, heavy machinery, and bulk safety gear. In the past, companies in this sector often tracked their sales using long text lists of numbers, which made it hard for managers to spot when sales were dipping or which channels were struggling. This project creates an easy-to-read, interactive dashboard that shows a full year of sales data in one centralized place, including online procurement portals and physical showrooms. The dashboard acts as a quick roadmap so a manager can instantly filter data by individual quarters, track seasonal revenue shifts, and decide where to send more inventory or shift sales staff.

Objectives

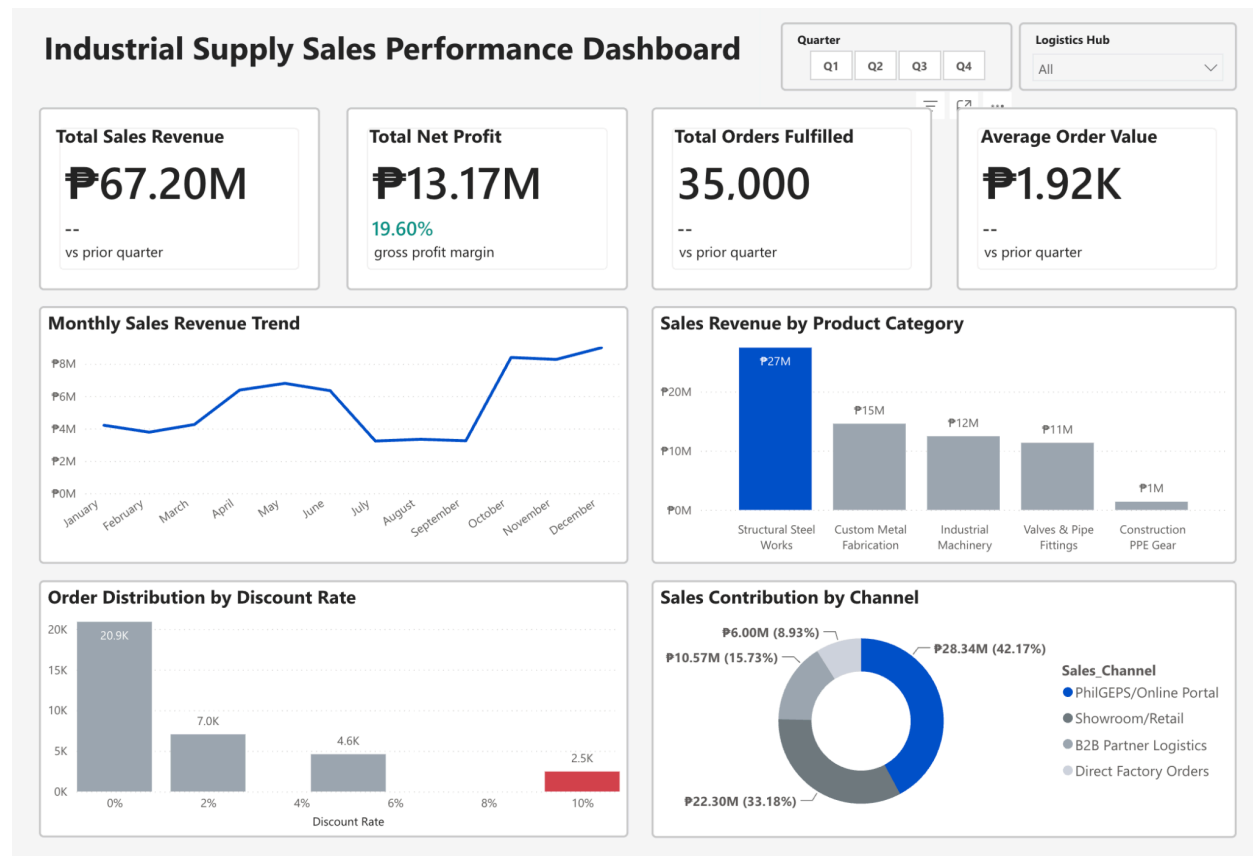
The final dashboard is built to hit five simple business goals across a full year of data:

- **Objective 1:** Show major annual numbers like total sales revenue, total net profit, unique order counts, and average order value in clean, uniform summary boxes at the very top.
- **Objective 2:** Draw a monthly trend line across all four quarters to help the manager compare busy seasons with natural sales drops, like the slow rainy season.
- **Objective 3:** Use a simple donut chart to split up sales by channel so we can instantly see if online bidding portals are beating physical showrooms year-round.
- **Objective 4:** Rank the main product lines from top to bottom so the manager can instantly spot the highest-earning items and flag slow categories that need help.
- **Objective 5:** Provide an interactive order distribution histogram to see exactly how frequently specific discount tiers are applied and find where aggressive pricing hurts final profit lines.

Dataset Description

- **Where the Data Came From:** The dataset was synthesized tracking 35,000 unique B2B transaction records generated using Python to simulate real world seasonal shifts in the industry.
- **What Fields Are Included:** The spreadsheet contains 14 clean columns including Transaction_ID, Date, Month, Quarter, Logistics_Hub, Client_Segment, Sales_Channel, Product_Category, Product_Item, Unit_Price_PHP, Quantity_Ordered, Discount_Rate, Net_Sales_PHP, and Net_Profit_PHP.
- **What Time Period it Covers:** The data covers a full 12-month calendar fiscal year, running continuously from **January 1 to December 31**.
- **How the Data Was Prepared:** We verified that our raw transaction data was completely clean, with zero spelling mistakes, blank fields, or duplicate rows. We then used Power Query to add a single numerical helper column (Month.1) so we could easily sort our text months in proper chronological order later on. Finally, we coded custom DAX measures (Total_Sales_Revenue, Total_Orders etc.) to handle all calculations live and used the Sort by Column tool to ensure our monthly trend line displayed a clean, seamless calendar timeline across all four quarters.

Dashboard Design Explanation



Why These Charts Were Chosen

- **KPI Summary Cards:** Four clean boxes at the very top give the manager instant answers to the most important business numbers (*Sales, Profit, Orders, and Average Order Value*).
- **Monthly Sales Trend (Line Chart):** Clearly shows how sales go up and down over a 12-month timeline so the manager can track seasonal demand.
- **Sales by Category (Column Chart):** Ranks the products from highest to lowest earner, showing the manager instantly what sells the most.

- **Sales Contribution (Donut Chart):** Shows the percentage share of each sales channel so the manager can track if online bidding is beating physical showrooms.
- **Order Distribution (Histogram):** Counts how many orders fall into each discount group to make sure sales reps aren't giving away too many price cuts.

How Layout and Grouping Help the Sales Manager (Gestalt Principles)

- **Proximity (Clustering):** Slicers and summary boxes are grouped closely at the top. This helps the manager because it acts like a unified control toolbar as they filter a region and instantly see the core numbers change right next to it.
- **Similarity (Matching Style):** Every chart uses the exact same fonts, borders, and currency symbols. This helps the manager read the dashboard faster because they do not have to waste time figuring out different styles as they switch charts.
- **Continuity (Natural Flow):** Charts are lined up in a neat, symmetrical grid. This lets the manager's eyes move naturally from left-to-right to connect *when* a trend happened (line chart) directly to *what product* caused it (column chart).
- **Figure-Ground (Making Data Pop):** Clean white chart boxes sit on top of a darker gray background frame. This forces the charts to stand out instantly so the manager is never distracted by the background wallpaper.
- **Readability:** Turning off messy gridlines and side titles lets the manager read exact numbers directly on top of the bars without any visual clutter blocking their view.

Color Logic

- **Blue Highlight:** Automatically colors *only* the number-one product and top sales channel. This cuts out noise and points the manager's eyes straight to their biggest revenue drivers.
- **Red Alert:** Used strictly as a warning color for danger zones. It instantly flags the high-risk 10% discount bar so the manager can stop profit loss immediately.
- **Muted Gray:** Keeps all secondary data bars calm and gray so they don't scream for attention or overwhelm the manager.

Key Insights

1. **The Q3 Rainy Season Slump:** When switching filters from Q2 to Q3, total sales revenue **drops by nearly half (from ₱19.51M to ₱9.81M)**. This clearly shows the impact of the Philippine monsoon season, which halts major outdoor construction projects nationwide.
2. **Rigid Warehouse Workloads:** Even though sales revenue cuts in half in Q3, the total number of orders stays flat at **8,695 in Q2 vs. 8,699 in Q3**. This means logistics hubs are packing the exact same number of trucks and handling the same paperwork, keeping warehouse operational costs high during slow months.
3. **Order Size Efficiency Loss:** Because order counts held steady while revenue dropped, the **Average Order Value plummeted from ₱2.24K to ₱1.13K**. In dry months, clients buy bulk materials; in rainy months, they buy small, emergency piecemeal replacements, which increases shipping costs per peso earned.

4. **Excellent Pricing Discipline:** Across all quarters and regional filters, the **Gross Profit Margin stays locked at 19.60%**. This proves that the sales team defended their markups and refused to give away reckless discounts to chase volume during the slow season.
5. **Online Portal Power:** The donut chart shows that the **PhilGEPS/Online Portal drives 42.17% (P28.34M)** of total revenue. Government bidding is the primary lifeblood of organization cash flow, while *Direct Factory Orders* are a small niche at just 8.93%.

Recommendations

1. **Set Q3 Minimum Order Sizes:** Since Q3 requires the same number of shipments as Q2 for half the money, introduce minimum order thresholds during rainy months to consolidate deliveries and cut truck fuel costs.
2. **Require Approval for 10% Discounts:** The histogram shows heavy transaction volume hitting the maximum **10% discount tier**. Management should require automatic manager sign-off for this specific tier to stop unnecessary margin leakage.
3. **Invest heavily in Online Bidding Teams:** Since the PhilGEPS online portal commands a massive 42.17% share, shift company budget away from expensive physical showrooms and invest directly into digital bidding automation and portal specialists.

Ethical Considerations

To ensure data honesty and prevent a misleading layout for the sales manager, the dashboard applies strict design guardrails:

- **Zero-Grounded Axes:** Vertical axes are locked to start at absolute zero. Charts are never cropped or stretched to visually exaggerate growth or manipulate seasonal declines.
- **Clutter Removal:** Redundant side axis titles were turned off. Data labels sit directly on top of the bars so the manager can read exact, real values without messy gridlines distorting the data scale.
- **Clear Suffix Text:** All percentages are paired with descriptive text labels (vs prior quarter, Gross Margin) so metrics are impossible to misinterpret or manipulate.

AI Usage Disclosure

AI was used as a technical collaborator to help generate the initial 35,000-row mock dataset using a structured Python loop, write the complex DAX formulas for the growth math and dynamic chart colors, and improve the wording throughout this report. However, all chart selections, visual layout design choices, and final data analysis interpretations were entirely decided and executed manually by the student. To ensure absolute data accuracy, all final Power BI dashboard numbers were cross-checked against the raw spreadsheet using Excel validation formulas like =SUM and =COUNTA. This manual check verified that the metrics matched the source data perfectly down to the single digit before final deployment.

Conclusion

The dashboard successfully turns a complex, 35,000-row industrial dataset into a clean, actionable corporate tool. For a sales manager, its primary value is exposing a major business reality, a heavy drop in shipping efficiency during the rainy season, balanced by strong institutional portal sales and excellent pricing discipline across regional hubs.

Through this project, I learned how to use Power Query to structure data columns, how to automate chart colors with DAX code to reduce a manager's cognitive load, and how to apply Gestalt principles to build an intuitive visual hierarchy. Most importantly, I learned that a great dashboard is made with intention and should never manipulate or exaggerate data, but rather clarify reality to drive smart corporate decisions.